SLIIT Department of Software Engineering - Faculty of Computing

Mobile Application Project: Farmora

Milestone 4: Tech Stack Justification & Mobile App Implementation (LO1, LO4)

Part 1: Tech Stack Justification

1. Architectural Options Comparison

Evaluation Criteria	Option A: Flutter (Selected)	Option B: React Native	Option C: Native (Kotlin + Swi
Rendering Performance	**High** (Skia/Impeller direct	**Medium** (JavaScript bridge	**Highest** (Native platform t
Multi-Platform Support	**Unified Codebase** (Android,	**High** (Mobile only; Web req	**None** (Requires 2 entirely
Team Skillset & Velocity	**Rapid** (Hot reload, typed D	**Moderate** (Requires reconci	**Slow** (Split team between S
Accessibility (a11y)	**Built-in** (Rich ‘Semantics‘	**Good** (Requires third-party	**High** (Native a11y APIs)
Hardware & Camera APIs	**Robust** (Direct Camera, QR	**Moderate** (Bridge lag can c	**High** (Full native access)
Development Cost	**Lowest** (1 codebase, 1 buil	**Moderate** (Bridge debugging	**Highest** (Requires distinct

2. Tech Stack Justification Rationale:

Flutter was selected because AgriTech users in Sri Lanka use diverse Android devices (from budget entry-level smartphones to modern flagships), alongside wholesale enterprise buyers accessing via desktop/tablet web browsers. Flutter provides:
1. **Identical Pixel-Perfect UI:** Zero layout discrepancies across different screen densities and aspect ratios.
2. **Offline-Ready Client Architecture:** Integrates seamlessly with Cloud Firestore offline caches and local SQLite/secure storage.
3. **High Productivity:** Hot Reload enabled the team to iterate rapidly during user testing in Milestone 3.

Part 2: Implementation & Folder Architecture

1. Folder Structure (Clean Architecture Pattern)

‘‘‘
lib/
??? core/
? ??? constants/ # AppColors, ApiEndpoints, Strings
? ??? network/ # Http & Cloudinary API clients
? ??? services/ # FirebaseAuthService, GeolocationService
? ??? theme/ # Material 3 light & dark themes
? ??? utils/ # Formatters, barcode helpers, image compression
??? features/
? ??? admin/ # Escrow dispute management, user approvals
? ??? auth/ # OTP verification, role switcher, onboarding

? ??? buyer/ # Catalog, video inspection, checkout, escrow
? ??? farmer/ # Harvest listing, disease detection, QR generator
? ??? transporter/ # Fleet Hub, depot manager, GPS tracking, POD
??? main.dart # Multi-provider initialization, route registry
‘‘‘

2. Version Control & Git Strategy

- Repository:** Hosted on GitHub (‘AHSANMOHAMMED/Farmora-App‘).
- Branching Model:** Feature-branch workflow with automated PR verification and strict branch protection on ‘main‘ and ‘dev/swami‘.
- Release Automation:** Continuous integration building automated Android APKs (‘app-release.apk‘) and continuous deployment to Firebase Web Hosting (‘farmingapp-24b34.web.app‘).

3. Requirements Traceability Matrix (RTM)

Req ID	Requirement / User Story	Implementation Screen / Widget	Data Model / Backend Endpoint	Status
REQ-01	Multi-role user registration	welcome_screen.dart‘, ‘role_	users/{uid}‘ (role, isVerifie	**Verified**
REQ-02	Farmers can record video‘ & add	product_screen.dart‘	crops/{cropId}‘ + Cloudinary	**Verified**
REQ-03	Generate unique batch QR for	batch_qr_dialog.dart‘, ‘qr_	lCodeHash‘, ‘storagePath‘	**Verified**
REQ-04	Buyers inspect video and ‘ place	oduct_detail_screen.dart‘,	rders/{orderId}‘ (escrowSta	**Verified**
REQ-05	Transporters manage branch logis	tics_fleet_hub_screen.da	nsporters/{tid}/vehicles‘,	**Verified**
REQ-06	Real-time Cold Storage telem	logistics_fleet_map_view.dart	old_storage_logs‘, ‘current	**Verified**
REQ-07	Proof-of-delivery (photo + ‘ sig	ider_dispatch_screen.dart	rders/{orderId}/podPhotoPath	**Verified**

4. Build Deliverables

- Mobile Build:** Android Release APK located at ‘build/app/outputs/flutter-apk/app-release.apk‘ (Size: 99.3MB, compileSdk: 36, Java 21 LTS).
- Web App URL:** https://farmingapp-24b34.web.app
- Git Repository:** https://github.com/AHSANMOHAMMED/Farmora-App.git